Probability - Example Sheet 1-2.

1.
a) How many ways are there to arrange 3 letters a, b, c? (E.g. $abc, acb, ...$)
b) How many permutations the 1, 2, 3, 4, 5 elements have, where the third digit is 1?
c) How many permutations the 1, 2, 3, 4, 5, 6, 7, 8 elements have, where 6, 7, 8 are in the first three places (in any order) and the last digit is 5?
d) How many odd 5-digit numbers can be made of 1, 2, 3, 4, 5 (without repetition)?
e) We have to seat 3 men and 4 women, so that men and women alternate. How many possible orders are there?

2.
a) How many ways are there to arrange 2 letters a and 2 letters b? (E.g. $aabb, abab, ...$)
b) How many permutations the 1, 2, 2, 3, 3, 3 elements have? How many of these have a digit 2 in the first place?
c) How many orders are possible for picking balls from an urn, if there are 5 white balls and 4 black balls in it? (We pick them all, and only the color makes a difference.)
d) How many orders are possible for picking balls from an urn, if there are 5 white balls, 4 black balls, and 3 red balls in it? (We pick them all, and only the color makes a difference.)

3.
a) How many words of 2 different letters can you make with 4 letters a, b, c, d? (E.g. $ab, ba, ac, ...$)
b) How many 4-digit numbers exist where the digits are all different?
c) How many 4-digit numbers exist where the digits are all different, and exactly two of the digits are odd?

4.
a) How many words of 2 letters can you make with 4 letters a, b, c, d? Repetition is allowed. (E.g. $aa, ab, ac, ad, ba, ...$)
b) How many 4-digit numbers exist, consisting of odd digits only?
c) How many 4-digit numbers exist, consisting of even digits only?
d) How many 5-digit numbers exist, consisting of 4 even and 1 odd digits? (Odd and even places are not fixed.)
e) In Hungary, the number plate of a car shows 3 letters and 3 digits (using 25 different letters and 10 digits). How many cars can be distinguished this way?

5.
a) How many ways are there to pick 2 different elements out of 5 elements 1, 2, 3, 4, 5? [Number of subsets.]
b) A company wants to hire 4 male and 4 female workers. 5 men and 8 women are applying for the positions. How many options are there to choose the workers?
c) From a pack of 52 cards, we give 8 cards to each of 3 players. How many different deals are possible?

6.
a) We have 2 (identical) red signalling balls, and 4 (numbered) green balls in a line. We can put the signalling balls in line with the green balls, showing our choice, putting them *before* a chosen ball. (Repeated choice is allowed.) How many possibilities we have for putting down the red balls among the green ones? (Hint: 5.a) helps.)
b) How many ways are there to pick 2 letters out of 4 letters? Repetition is allowed, and the order does not matter. (E.g. $aa, ab, ac, ad, bb...$)
c) How many ways are there to pick 5 letters out of 27 letters? Repetition is allowed, and the order does not matter.

7.
Six cups and saucers come in pairs: there are two cups and saucers which are red, two white, and two with stars on. If the cups are placed randomly onto the saucers (one each), find the probability that no cup is upon a saucer of the same pattern.

8.
We roll a die n times. Let A_{ij} be the event that the ith and jth rolls produce the same number. Show that the events $A_{ij}: \ 1 \leq i < j \leq n$ are pairwise independent but not independent.

9.
a) A man possesses five coins, two of which are double-headed, one is double-tailed, and two are normal. He shuts his eyes, picks a coin at random, and tosses it. What is the probability that the lower face of the coin is a head?
b) He opens his eyes and sees that the coin is showing heads; what is the probability that the lower face is a head?
c) He shuts his eyes again, and tosses the coin again. What is the probability that the lower face is a head?
d) He opens his eyes and sees that the coin is showing heads; what is the probability that the lower face is a head?
e) He discards this coin, picks another at random, and tosses it. What is the probability that it shows heads?

10.
Let $A_1, A_2, ..., A_n$ be events where $n \geq 2$, and prove the *inclusion-exclusion principle* (also known as *sieve principle*):

$$P(\bigcup_{i=1}^{n} A_i) = \sum_i P(A_i) - \sum_{i<j} P(A_i \cap A_j) + \sum_{i<j<k} P(A_i \cap A_j \cap A_k) - ... + (-1)^{n+1} P(A_1 \cap A_2 \cap ... \cap A_n).$$

Apply this formula for the following question.
In each packet of Corn Flakes may be found a plastic bust of one of the last five Vice-Chancellors of Cambridge University, the probability that any given packet contains any specific Vice-Chancellor being $\frac{1}{5}$, independently of all other packets. Show that the probability that each of the last three Vice-Chancellors is obtained in a bulk purchase of six packets is $1 - 3(\frac{4}{5})^6 + 3(\frac{3}{5})^6 - (\frac{2}{5})^6$.